

20XD57 - MACHINE LEARNING LAB

PACKAGE ABSTRACT

Topic :

Comparing clustering methods on investors dataset.

Team :

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Dataset :

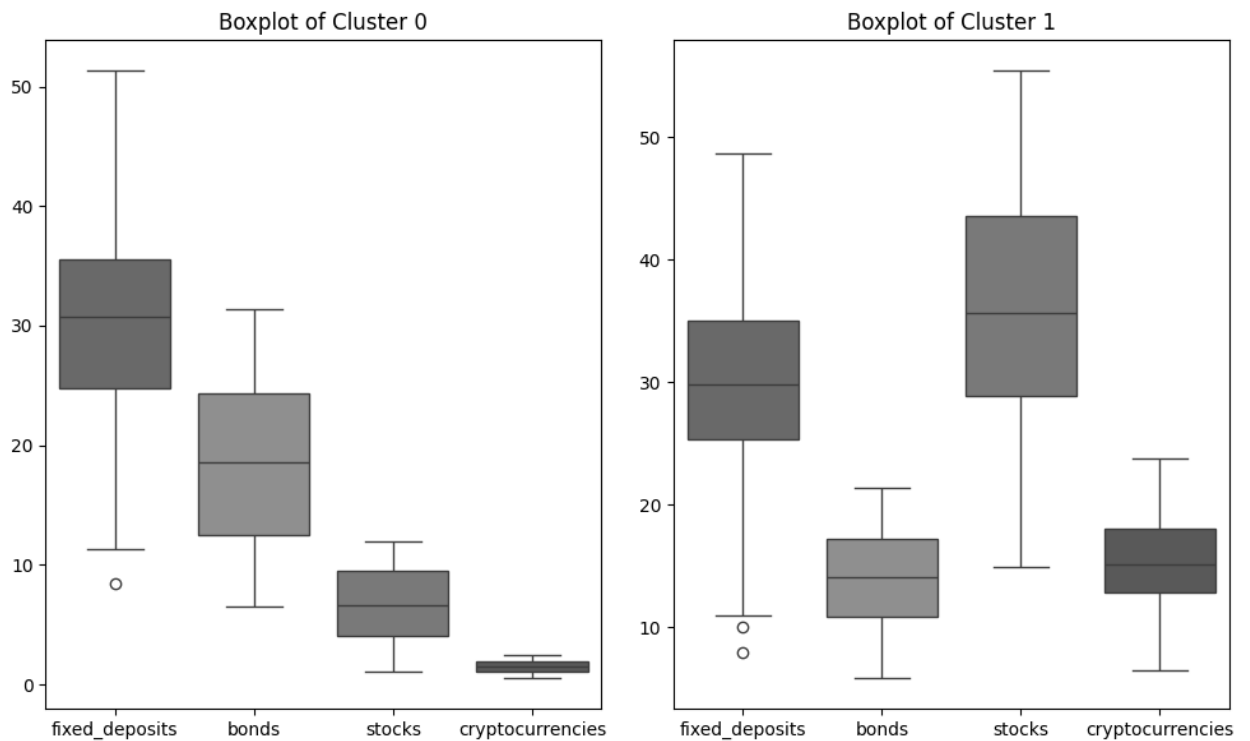
- Features(4) :
 - fixed_deposits
 - bonds
 - stocks
 - cryptocurrencies
- Classes(2) :
 - 0 - low-risk investors
 - 1 - high-risk investors

Summary :

- Investors dataset without class column is taken for clustering.
- On plotting the inertia vs k (elbow plot) we find optimal no. of clusters $k = 2$, which corresponds to no. of classes.
- On applying k-means clustering from the sklearn library and comparing the clusters with the actual classes, we get an accuracy of 99%.
- On applying spectral clustering from the sklearn library and comparing the clusters with the actual classes, we get an accuracy score of 100%.
- On manually applying spectral clustering, we get the same accuracy that we get when using libraries.

Inference :

- On plotting the boxplot of the two clusters, we can see significant differences in the amount of investments in the stocks and cryptocurrencies asset classes.
- Cluster 1 has more allocation in those asset classes, while cluster 0 has very less allocation in the same.



Conclusion :

- Both k-means clustering and spectral clustering were able to successfully cluster the dataset.
- There is no significant difference between the performance of k-means clustering and spectral clustering on the dataset.